

RockfallGuard AI Project Guide

Easy & Simple Explanation of the System

What is RockfallGuard?

RockfallGuard is a smart AI safety application built for open-pit mines. It monitors sensors in real time and predicts dangerous rock collapses **BEFORE** they happen, helping evacuate workers safely and save lives.

1. The Problem We Are Solving

Open-pit mines are giant open-air holes in the earth where metals like copper and gold are mined. The steep walls of these pits can collapse without warning due to heavy rainfall or underground water pressure. Older safety systems only ring an alarm after the ground starts falling, which is too late! RockfallGuard uses AI to give an **early warning**.

2. How RockfallGuard Works (5 Simple Steps)

Step 1: Sensors in the Ground — Measures ground movement, underground water pressure, and micro-cracking vibrations.

Step 2: Live Weather Telemetry — Uses the mine's exact GPS location to check live rainfall and humidity.

Step 3: Noise Filtering — Uses a smart digital filter to remove noise caused by heavy mining trucks driving by, preventing false alarms.

Step 4: AI Risk Score (99.5% Accurate) — The AI calculates a live 0–100% Failure Risk Score:

- **GREEN (Safe <35%)**: Normal safe mining operations.
- **YELLOW (Warning 35–65%)**: Caution required; inspect bench slope.
- **RED (Critical >65%)**: DANGER! Evacuate bench sectors immediately!

Step 5: Automated Email & SMS Alerts — If risk goes above 80%, the system automatically sounds a loud siren, flashes red warnings, and sends emergency Emails and SMS text messages to mine operators.

3. Main Features Included in the Web Application

Feature Name	What It Does
■ Live Hazard Risk Gauge	Displays continuous 0-100% failure risk in real time.
■ SHAP AI Root Cause Explainer	tells you WHY the risk went up (e.g. 'High Rainfall' or 'Ground Acceleration').
■ Drone Photo Crack Inspection	Upload photos taken by aerial drones to let PyTorch AI scan for surface rock cracks.
■■ Interactive Mine Location Map	Shows a live interactive satellite map pin of where the mine is located on Earth.
■ Register & Delete Mines	Easily add new mining pits or delete old ones from the database.
■ Emergency Email & SMS Alerts	Automatically alerts safety officers via Email and SMS when danger exceeds 80%.

4. How to Use the Dashboard

1. Open **http://127.0.0.1:8009** in your web browser.
2. Select your active mine from the dropdown menu at the top.
3. Use the simulator buttons to test scenarios (Normal, Rain, Machinery Noise, Emergency Alert).
4. Click **■ Drone CNN Inspection** to upload and inspect drone photos.
5. Click **■■ Delete** if you want to remove a mine from the system.